

Square Roots of Perfect Squares

Jonathan R. Bacolod

A **square root** of a number is a value that, when multiplied by itself, gives the original number.

Perfect Square

A **perfect square** is a number that can be written as the square of a whole number.

$$25 = 5^2$$

Therefore, 25 is a perfect square.

Radical Sign

The symbol $\sqrt{\quad}$ is called the **radical sign**.

$$\sqrt{25} = 5$$

because

$$5 \times 5 = 25$$

Finding Square Roots

To determine the square root of a perfect square, find the number that, when multiplied by itself, equals the given number.

$$\sqrt{64} = 8$$

because

$$8 \times 8 = 64$$

Principal Square Root

The **principal square root** is the positive square root of a number.

$$\sqrt{121} = 11$$

The principal square root is 11.

Examples

Example 1

Find $\sqrt{25}$.

$$\sqrt{25} = 5$$

because

$$5 \times 5 = 25$$

Example 2

Find $\sqrt{64}$.

$$\sqrt{64} = 8$$

because

$$8 \times 8 = 64$$

Example 3

Find $\sqrt{121}$.

$$\sqrt{121} = 11$$

because

$$11 \times 11 = 121$$

Practice Exercises

Practice 1

Find the square root.

$$\sqrt{16}$$

Practice 2

Find the square root.

$$\sqrt{49}$$

Practice 3

Find the square root.

$$\sqrt{81}$$

Practice 4

Find the square root.

$$\sqrt{100}$$

Practice 5

Find the square root.

$$\sqrt{144}$$

Practice 6

Find the square root.

$$\sqrt{225}$$

Practice 7

Find the square root.

$$\sqrt{324}$$

Practice 8

Find the square root.

$$\sqrt{400}$$

Practice 9

Find the square root.

$$\sqrt{625}$$

Practice 10

Find the square root.

$$\sqrt{900}$$

Activities

Activity 1

Find the square root.

$$\sqrt{9}$$

Activity 2

Find the square root.

$$\sqrt{36}$$

Activity 3

Find the square root.

$$\sqrt{64}$$

Activity 4

Find the square root.

$$\sqrt{121}$$

Activity 5

Find the square root.

$$\sqrt{169}$$

Activity 6

Find the square root.

$$\sqrt{196}$$

Activity 7

Find the square root.

$$\sqrt{289}$$

Activity 8

Find the square root.

$$\sqrt{441}$$

Activity 9

Find the square root.

$$\sqrt{784}$$

Activity 10

Find the square root.

$$\sqrt{1024}$$

Thank you!